

## 10.4 Nonhomogeneous Linear Systems

**Introduction** The methods of **undetermined coefficients** and **variation of parameters** used in Chapter 3 to find particular solutions of nonhomogeneous linear ODEs can both be adapted to the solution of nonhomogeneous linear systems. Of the two methods, variation of parameters is the more powerful technique. However, there are instances when the method of undetermined coefficients provides a quick means of finding a particular solution.

In Section 10.1, we saw that the general solution of a nonhomogeneous linear system  $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t)$  on an interval  $I$  is  $\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$  where  $\mathbf{X}_c = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \cdots + c_n\mathbf{X}_n$  is the complementary function or general solution of the associated homogeneous linear system  $\mathbf{X}' = \mathbf{A}\mathbf{X}$  and  $\mathbf{X}_p$  is any particular solution of the nonhomogeneous system. We just saw how to obtain  $\mathbf{X}_c$  in Section 10.2 when  $\mathbf{A}$  was an  $n \times n$  matrix of constants; we now consider three methods for obtaining  $\mathbf{X}_p$ .

### 10.4.1 Undetermined Coefficients

**The Assumptions** As in Section 3.4, the **method of undetermined coefficients** consists of making an educated guess about the form of a particular solution vector  $\mathbf{X}_p$ ; the guess is motivated by the types of functions that comprise the entries of the column matrix  $\mathbf{F}(t)$ . Not surprisingly, the matrix version of undetermined coefficients is only applicable to  $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t)$  when the entries of  $\mathbf{A}$  are constants and the entries of  $\mathbf{F}(t)$  are constants, polynomials, exponential functions, sines and cosines, or finite sums and products of these functions.

#### EXAMPLE 1 Undetermined Coefficients

Solve the system  $\mathbf{X}' = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} -8 \\ 3 \end{pmatrix}$  on the interval  $(-\infty, \infty)$ .

**SOLUTION** We first solve the associated homogeneous system

$$\mathbf{X}' = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \mathbf{X}.$$

The characteristic equation of the coefficient matrix  $\mathbf{A}$ ,

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -1 - \lambda & 2 \\ -1 & 1 - \lambda \end{vmatrix} = \lambda^2 + 1 = 0,$$

yields the complex eigenvalues  $\lambda_1 = i$  and  $\lambda_2 = \bar{\lambda}_1 = -i$ . By the procedures of Section 10.2, we find

$$\mathbf{X}_c = c_1 \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix} + c_2 \begin{pmatrix} \cos t - \sin t \\ -\sin t \end{pmatrix}.$$

Now since  $\mathbf{F}(t)$  is a constant vector, we assume a constant particular solution vector  $\mathbf{X}_p = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$ .

Substituting this latter assumption into the original system and equating entries leads to

$$\begin{aligned} 0 &= -a_1 + 2b_1 - 8 \\ 0 &= -a_1 + b_1 + 3. \end{aligned}$$

Solving this algebraic system gives  $a_1 = 14$  and  $b_1 = 11$ , and so a particular solution is  $\mathbf{X}_p = \begin{pmatrix} 14 \\ 11 \end{pmatrix}$ . The general solution of the original system of DEs on  $(-\infty, \infty)$  is then  $\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$  or

$$\mathbf{X} = c_1 \begin{pmatrix} \cos t + \sin t \\ \cos t \end{pmatrix} + c_2 \begin{pmatrix} \cos t - \sin t \\ -\sin t \end{pmatrix} + \begin{pmatrix} 14 \\ 11 \end{pmatrix}. \quad \equiv$$

**EXAMPLE 2** Undetermined Coefficients

Solve the system  $\mathbf{X}' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 6t \\ -10t + 4 \end{pmatrix}$  on the interval  $(-\infty, \infty)$ .

**SOLUTION** The eigenvalues and corresponding eigenvectors of the associated homogeneous system  $\mathbf{X}' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} \mathbf{X}$  are found to be  $\lambda_1 = 2$ ,  $\lambda_2 = 7$ ,  $\mathbf{K}_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$ , and  $\mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ . Hence the complementary function is

$$\mathbf{X}_c = c_1 \begin{pmatrix} 1 \\ -4 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{7t}.$$

Now because  $\mathbf{F}(t)$  can be written  $\mathbf{F}(t) = \begin{pmatrix} 6 \\ -10 \end{pmatrix} t + \begin{pmatrix} 0 \\ 4 \end{pmatrix}$  we shall try to find a particular solution of the system that possesses the *same* form:

$$\mathbf{X}_p = \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} t + \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}.$$

Substituting this last assumption into the given system yields

$$\begin{pmatrix} a_2 \\ b_2 \end{pmatrix} = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} \left[ \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} t + \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} \right] + \begin{pmatrix} 6 \\ -10 \end{pmatrix} t + \begin{pmatrix} 0 \\ 4 \end{pmatrix}$$

or

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} (6a_2 + b_2 + 6)t + 6a_1 + b_1 - a_2 \\ (4a_2 + 3b_2 - 10)t + 4a_1 + 3b_1 - b_2 + 4 \end{pmatrix}.$$

From the last identity we obtain four algebraic equations in four unknowns:

$$\begin{aligned} 6a_2 + b_2 + 6 &= 0 & \text{and} & & 6a_1 + b_1 - a_2 &= 0 \\ 4a_2 + 3b_2 - 10 &= 0 & & & 4a_1 + 3b_1 - b_2 + 4 &= 0. \end{aligned}$$

Solving the first two equations simultaneously yields  $a_2 = -2$  and  $b_2 = 6$ . We then substitute these values into the last two equations and solve for  $a_1$  and  $b_1$ . The results are  $a_1 = -\frac{4}{7}$ ,  $b_1 = \frac{10}{7}$ . It follows, therefore, that a particular solution vector is

$$\mathbf{X}_p = \begin{pmatrix} -2 \\ 6 \end{pmatrix} t + \begin{pmatrix} -\frac{4}{7} \\ \frac{10}{7} \end{pmatrix}.$$

The general solution of the system on  $(-\infty, \infty)$  is  $\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$  or

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ -4 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{7t} + \begin{pmatrix} -2 \\ 6 \end{pmatrix} t + \begin{pmatrix} -\frac{4}{7} \\ \frac{10}{7} \end{pmatrix}. \quad \equiv$$

**EXAMPLE 3** Form of  $\mathbf{X}_p$ 

Determine the form of a particular solution vector  $\mathbf{X}_p$  for the system

$$\begin{aligned} \frac{dx}{dt} &= 5x + 3y - 2e^{-t} + 1 \\ \frac{dy}{dt} &= -x + y + e^{-t} - 5t + 7. \end{aligned}$$

**SOLUTION** Because  $\mathbf{F}(t)$  can be written in matrix terms as

$$\mathbf{F}(t) = \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ -5 \end{pmatrix} t + \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

a natural assumption for a particular solution would be

$$\mathbf{X}_p = \begin{pmatrix} a_3 \\ b_3 \end{pmatrix} e^{-t} + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} t + \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}. \quad \equiv$$

## Remarks

The method of undetermined coefficients for linear systems is not as straightforward as the last three examples would seem to indicate. In Section 3.4, the form of a particular solution  $y_p$  was predicated on prior knowledge of the complementary function  $y_c$ . The same is true for the formation of  $\mathbf{X}_p$ . But there are further difficulties; the special rules governing the form of  $y_p$  in Section 3.4 do not *quite* carry to the formation of  $\mathbf{X}_p$ . For example, if  $\mathbf{F}(t)$  is a constant vector as in Example 1 and  $\lambda = 0$  is an eigenvalue of multiplicity one, then  $\mathbf{X}_c$  contains a constant vector. Under the “multiplication rule” on page 130, we would ordinarily try a particular solution of the form  $\mathbf{X}_p = \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} t$ . This is not the proper assumption for linear systems; it should be  $\mathbf{X}_p = \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} t + \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$ . Similarly, in Example 3, if we replace  $e^{-t}$  in  $\mathbf{F}(t)$  by  $e^{2t}$  ( $\lambda = 2$  is an eigenvalue), then the correct form of the particular solution vector is

$$\mathbf{X}_p = \begin{pmatrix} a_4 \\ b_4 \end{pmatrix} t e^{2t} + \begin{pmatrix} a_3 \\ b_3 \end{pmatrix} e^{2t} + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} t + \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}.$$

Rather than delve into these difficulties, we turn instead to the method of variation of parameters.

## 10.4.2 Variation of Parameters

**A Fundamental Matrix** If  $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$  is a fundamental set of solutions of the homogeneous system  $\mathbf{X}' = \mathbf{A}\mathbf{X}$  on an interval  $I$ , then its general solution on the interval is the linear combination  $\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2 + \dots + c_n\mathbf{X}_n$ , or

$$\mathbf{X} = c_1 \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix} + c_2 \begin{pmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{pmatrix} + \dots + c_n \begin{pmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{pmatrix} = \begin{pmatrix} c_1x_{11} + c_2x_{12} + \dots + c_nx_{1n} \\ c_1x_{21} + c_2x_{22} + \dots + c_nx_{2n} \\ \vdots \\ c_1x_{n1} + c_2x_{n2} + \dots + c_nx_{nn} \end{pmatrix}. \quad (1)$$

The last matrix in (1) is recognized as the product of an  $n \times n$  matrix with an  $n \times 1$  matrix. In other words, the general solution (1) can be written as the product

$$\mathbf{X} = \Phi(t)\mathbf{C}, \quad (2)$$

where  $\mathbf{C}$  is the  $n \times 1$  column vector of arbitrary constants  $c_1, c_2, \dots, c_n$ , and  $\Phi(t)$  is the  $n \times n$  matrix whose columns consist of the entries of the solution vectors of the system  $\mathbf{X}' = \mathbf{A}\mathbf{X}$ :

$$\Phi(t) = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{pmatrix}.$$

The matrix  $\Phi(t)$  is called a **fundamental matrix** of the system on the interval.

In the discussion that follows, we need to use two properties of a fundamental matrix:

- A fundamental matrix  $\Phi(t)$  is nonsingular.
- If  $\Phi(t)$  is a fundamental matrix of the system  $\mathbf{X}' = \mathbf{A}\mathbf{X}$ , then

$$\Phi'(t) = \mathbf{A}\Phi(t). \quad (3)$$

A reexamination of (9) of Theorem 10.1.3 shows that  $\det \Phi(t)$  is the same as the Wronskian  $W(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n)$ . Hence the linear independence of the columns of  $\Phi(t)$  on the interval  $I$  guarantees that  $\det \Phi(t) \neq 0$  for every  $t$  in the interval. Since  $\Phi(t)$  is nonsingular, the multiplicative inverse  $\Phi^{-1}(t)$  exists for every  $t$  in the interval. The result given in (3) follows immediately from the fact that every column of  $\Phi(t)$  is a solution vector of  $\mathbf{X}' = \mathbf{A}\mathbf{X}$ .

**□ Variation of Parameters** Analogous to the procedure in Section 3.5, we ask whether it is possible to replace the matrix of constants  $\mathbf{C}$  in (2) by a column matrix of functions

$$\mathbf{U}(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_n(t) \end{pmatrix} \quad \text{so that} \quad \mathbf{X}_p = \Phi(t)\mathbf{U}(t) \quad (4)$$

is a particular solution of the nonhomogeneous system

$$\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t). \quad (5)$$

By the Product Rule, the derivative of the last expression in (4) is

$$\mathbf{X}'_p = \Phi(t)\mathbf{U}'(t) + \Phi'(t)\mathbf{U}(t). \quad (6)$$

Note that the order of the products in (6) is very important. Since  $\mathbf{U}(t)$  is a column matrix, the products  $\mathbf{U}'(t)\Phi(t)$  and  $\mathbf{U}(t)\Phi'(t)$  are not defined. Substituting (4) and (6) into (5) gives

$$\Phi(t)\mathbf{U}'(t) + \Phi'(t)\mathbf{U}(t) = \mathbf{A}\Phi(t)\mathbf{U}(t) + \mathbf{F}(t). \quad (7)$$

Now if we use (3) to replace  $\Phi'(t)$ , (7) becomes

$$\Phi(t)\mathbf{U}'(t) + \mathbf{A}\Phi(t)\mathbf{U}(t) = \mathbf{A}\Phi(t)\mathbf{U}(t) + \mathbf{F}(t)$$

or

$$\Phi(t)\mathbf{U}'(t) = \mathbf{F}(t). \quad (8)$$

Multiplying both sides of equation (8) by  $\Phi^{-1}(t)$  gives

$$\mathbf{U}'(t) = \Phi^{-1}(t)\mathbf{F}(t) \quad \text{and so} \quad \mathbf{U}(t) = \int \Phi^{-1}(t)\mathbf{F}(t) dt.$$

Since  $\mathbf{X}_p = \Phi(t)\mathbf{U}(t)$ , we conclude that a particular solution of (5) is

$$\mathbf{X}_p = \Phi(t) \int \Phi^{-1}(t)\mathbf{F}(t) dt. \quad (9)$$

To calculate the indefinite integral of the column matrix  $\Phi^{-1}(t)\mathbf{F}(t)$  in (9), we integrate each entry. Thus the general solution of the system (5) is  $\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$  or

$$\mathbf{X} = \Phi(t)\mathbf{C} + \Phi(t) \int \Phi^{-1}(t)\mathbf{F}(t) dt. \quad (10)$$

#### EXAMPLE 4 Variation of Parameters

Find the general solution of the nonhomogeneous system

$$\mathbf{X}' = \begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3t \\ e^{-t} \end{pmatrix} \quad (11)$$

on the interval  $(-\infty, \infty)$ .

**SOLUTION** We first solve the associated homogeneous system

$$\mathbf{X}' = \begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix} \mathbf{X}. \quad (12)$$

The characteristic equation of the coefficient matrix is

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -3 - \lambda & 1 \\ 2 & -4 - \lambda \end{vmatrix} = (\lambda + 2)(\lambda + 5) = 0,$$

so the eigenvalues are  $\lambda_1 = -2$  and  $\lambda_2 = -5$ . By the usual method we find that the eigenvectors corresponding to  $\lambda_1$  and  $\lambda_2$  are, respectively,

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

The solution vectors of the homogeneous system (12) are then

$$\mathbf{X}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} = \begin{pmatrix} e^{-2t} \\ e^{-2t} \end{pmatrix} \quad \text{and} \quad \mathbf{X}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-5t} = \begin{pmatrix} e^{-5t} \\ -2e^{-5t} \end{pmatrix}.$$

The entries in  $\mathbf{X}_1$  form the first column of  $\Phi(t)$ , and the entries in  $\mathbf{X}_2$  form the second column of  $\Phi(t)$ . Hence

$$\Phi(t) = \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix} \quad \text{and} \quad \Phi^{-1}(t) = \begin{pmatrix} \frac{2}{3}e^{2t} & \frac{1}{3}e^{2t} \\ \frac{1}{3}e^{5t} & -\frac{1}{3}e^{5t} \end{pmatrix}.$$

From (9) we obtain

$$\begin{aligned} \mathbf{X}_p &= \Phi(t) \int \Phi^{-1}(t) \mathbf{F}(t) dt = \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix} \int \begin{pmatrix} \frac{2}{3}e^{2t} & \frac{1}{3}e^{2t} \\ \frac{1}{3}e^{5t} & -\frac{1}{3}e^{5t} \end{pmatrix} \begin{pmatrix} 3t \\ e^{-t} \end{pmatrix} dt \\ &= \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix} \int \begin{pmatrix} 2te^{2t} + \frac{1}{3}e^t \\ te^{5t} - \frac{1}{3}e^{4t} \end{pmatrix} dt \\ &= \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix} \begin{pmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{1}{5}te^{5t} - \frac{1}{25}e^{5t} - \frac{1}{12}e^{4t} \end{pmatrix} \\ &= \begin{pmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{3}{5}t - \frac{21}{50} + \frac{1}{2}e^{-t} \end{pmatrix}. \end{aligned}$$

Hence from (10), the general solution of (11) on the interval is

$$\begin{aligned} \mathbf{X} &= \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{3}{5}t - \frac{21}{50} + \frac{1}{2}e^{-t} \end{pmatrix} \\ &= c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-5t} + \begin{pmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{3}{5}t - \frac{21}{50} + \frac{1}{2}e^{-t} \end{pmatrix}. \quad \equiv \end{aligned}$$

**Initial-Value Problem** The general solution of the nonhomogeneous system (5) on an interval can be written in an alternative manner

$$\mathbf{X} = \Phi(t)\mathbf{C} + \Phi(t) \int_{t_0}^t \Phi^{-1}(s)\mathbf{F}(s) ds, \quad (13)$$

where  $t$  and  $t_0$  are points in the interval. The last form is useful in solving (5) subject to an initial condition  $\mathbf{X}(t_0) = \mathbf{X}_0$ , because the limits of integration are chosen so that the particular solution

vanishes at  $t = t_0$ . Substituting  $t = t_0$  in (13) yields  $\mathbf{X}_0 = \Phi(t_0)\mathbf{C}$ , from which we get  $\mathbf{C} = \Phi^{-1}(t_0)\mathbf{X}_0$ . Substituting this last result in (13) gives the following solution of the initial-value problem:

$$\mathbf{X} = \Phi(t)\Phi^{-1}(t_0)\mathbf{X}_0 + \Phi(t)\int_{t_0}^t \Phi^{-1}(s)\mathbf{F}(s) ds. \quad (14)$$

### 10.4.3 Diagonalization

**The Assumptions** As in Section 10.3, if the coefficient matrix  $\mathbf{A}$  possesses  $n$  linearly independent eigenvectors, then we can use diagonalization to uncouple the system  $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t)$ . Suppose  $\mathbf{P}$  is the matrix such that  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$ , where  $\mathbf{D}$  is a diagonal matrix. Substituting  $\mathbf{X} = \mathbf{P}\mathbf{Y}$  into the nonhomogeneous system  $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t)$  gives

$$\mathbf{P}\mathbf{Y}' = \mathbf{A}\mathbf{P}\mathbf{Y} + \mathbf{F} \quad \text{or} \quad \mathbf{Y}' = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{Y} + \mathbf{P}^{-1}\mathbf{F} \quad \text{or} \quad \mathbf{Y}' = \mathbf{D}\mathbf{Y} + \mathbf{G}. \quad (15)$$

In the last equation in (15),  $\mathbf{G} = \mathbf{P}^{-1}\mathbf{F}$  is a column vector. So each differential equation in this new system has the form  $y_i' = \lambda_i y_i + g_i(t)$ ,  $i = 1, 2, \dots, n$ . But notice that, unlike the procedure for solving a homogeneous system  $\mathbf{X}' = \mathbf{A}\mathbf{X}$ , we now are required to compute the inverse of the matrix  $\mathbf{P}$ .

#### EXAMPLE 5 Diagonalization

Solve  $\mathbf{X}' = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3e^t \\ e^t \end{pmatrix}$  by diagonalization.

**SOLUTION** The eigenvalues and corresponding eigenvectors of the coefficient matrix are found to be  $\lambda_1 = 0$ ,  $\lambda_2 = 5$ ,  $\mathbf{K}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ ,  $\mathbf{K}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ . Thus we find  $\mathbf{P} = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$  and  $\mathbf{P}^{-1} = \begin{pmatrix} \frac{1}{5} & -\frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{pmatrix}$ . Using the substitution  $\mathbf{X} = \mathbf{P}\mathbf{Y}$  and

$$\mathbf{P}^{-1}\mathbf{F} = \begin{pmatrix} \frac{1}{5} & -\frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{pmatrix} \begin{pmatrix} 3e^t \\ e^t \end{pmatrix} = \begin{pmatrix} \frac{1}{5}e^t \\ \frac{7}{5}e^t \end{pmatrix}$$

the uncoupled system is

$$\mathbf{Y}' = \begin{pmatrix} 0 & 0 \\ 0 & 5 \end{pmatrix} \mathbf{Y} + \begin{pmatrix} \frac{1}{5}e^t \\ \frac{7}{5}e^t \end{pmatrix}.$$

The solutions of the two linear first-order differential equations

$$y_1' = \frac{1}{5}e^t \quad \text{and} \quad y_2' = 5y_2 + \frac{7}{5}e^t$$

are, respectively,  $y_1 = \frac{1}{5}e^t + c_1$  and  $y_2 = -\frac{7}{20}e^t + c_2e^{5t}$ . Hence, the solution of the original system is

$$\mathbf{X} = \mathbf{P}\mathbf{Y} = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{5}e^t + c_1 \\ -\frac{7}{20}e^t + c_2e^{5t} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2}e^t + c_1 + 2c_2e^{5t} \\ -\frac{3}{4}e^t - 2c_1 + c_2e^{5t} \end{pmatrix}. \quad (16)$$

Written in the usual manner using column vectors, (16) is

$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ -2 \end{pmatrix} + c_2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} e^{5t} - \begin{pmatrix} \frac{1}{2} \\ \frac{3}{4} \end{pmatrix} e^t. \quad \equiv$$

## 10.4 Exercises

Answers to selected odd-numbered problems begin on page ANS-26.

### 10.4.1 Undetermined Coefficients

In Problems 1–8, use the method of undetermined coefficients to solve the given system.

$$1. \begin{cases} \frac{dx}{dt} = 2x + 3y - 7 \\ \frac{dy}{dt} = -x - 2y + 5 \end{cases}$$

$$2. \begin{cases} \frac{dx}{dt} = 5x + 9y + 2 \\ \frac{dy}{dt} = -x + 11y + 6 \end{cases}$$

$$3. \mathbf{X}' = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} -2t^2 \\ t + 5 \end{pmatrix}$$

$$4. \mathbf{X}' = \begin{pmatrix} 1 & -4 \\ 4 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 4t + 9e^{6t} \\ -t + e^{6t} \end{pmatrix}$$

$$5. \mathbf{X}' = \begin{pmatrix} 4 & \frac{1}{3} \\ 9 & 6 \end{pmatrix} \mathbf{X} + \begin{pmatrix} -3 \\ 10 \end{pmatrix} e^t$$